

```
mysql> SELECT * FROM customers LIMIT 5;
```

| customer_id | first_name | last_name | email              | city      | state       | join_date  | is_premium |
|-------------|------------|-----------|--------------------|-----------|-------------|------------|------------|
| 101         | Aarav      | Sharma    | aarav.s@email.com  | Mumbai    | Maharashtra | 2024-01-15 | 1          |
| 102         | Priya      | Patel     | priya.p@email.com  | Ahmedabad | Gujarat     | 2024-02-20 | 0          |
| 103         | Rohan      | Gupta     | rohan.g@email.com  | Delhi     | Delhi       | 2024-03-10 | 1          |
| 104         | Sneha      | Reddy     | sneha.r@email.com  | Hyderabad | Telangana   | 2024-04-05 | 0          |
| 105         | Vikram     | Singh     | vikram.s@email.com | Jaipur    | Rajasthan   | 2024-05-12 | 1          |

5 rows in set (0.00 sec)

```
mysql> SELECT * FROM orders LIMIT 5;
```

| order_id | customer_id | order_date | status    | total_amount |
|----------|-------------|------------|-----------|--------------|
| 1001     | 101         | 2024-08-01 | Delivered | 4498.00      |
| 1002     | 102         | 2024-08-03 | Delivered | 799.00       |
| 1003     | 103         | 2024-08-05 | Shipped   | 7498.00      |
| 1004     | 101         | 2024-08-10 | Delivered | 3499.00      |
| 1005     | 104         | 2024-08-12 | Cancelled | 2999.00      |

5 rows in set (0.00 sec)

```
mysql> SELECT * FROM products LIMIT 5;
```

| product_id | product_name      | category    | brand | unit_price | stock_qty |
|------------|-------------------|-------------|-------|------------|-----------|
| 201        | Wireless Earbuds  | Electronics | BoAt  | 1499.00    | 250       |
| 202        | Cotton T-Shirt    | Clothing    | Levis | 799.00     | 500       |
| 203        | Smart Watch       | Electronics | Noise | 2999.00    | 150       |
| 204        | Running Shoes     | Clothing    | Nike  | 4599.00    | 120       |
| 205        | Bluetooth Speaker | Electronics | JBL   | 3499.00    | 200       |

5 rows in set (0.00 sec)

```
mysql> SELECT * FROM order_items LIMIT 5;
```

| item_id | order_id | product_id | quantity | unit_price | discount_pct |
|---------|----------|------------|----------|------------|--------------|
| 5001    | 1001     | 201        | 2        | 1499.00    | 0.00         |
| 5002    | 1001     | 207        | 1        | 899.00     | 10.00        |
| 5003    | 1002     | 202        | 1        | 799.00     | 0.00         |
| 5004    | 1003     | 203        | 1        | 2999.00    | 0.00         |
| 5005    | 1003     | 204        | 1        | 4599.00    | 5.00         |

5 rows in set (0.00 sec)

```
mysql> SELECT * FROM customers WHERE STATE="MAHARASHTRA";
```

| customer_id | first_name | last_name | email             | city   | state       | join_date  | is_premium |
|-------------|------------|-----------|-------------------|--------|-------------|------------|------------|
| 101         | Aarav      | Sharma    | aarav.s@email.com | Mumbai | Maharashtra | 2024-01-15 | 1          |
| 107         | Karan      | Mehta     | karan.m@email.com | Pune   | Maharashtra | 2024-07-22 | 1          |

2 rows in set (0.02 sec)

```
mysql> SELECT * FROM customers WHERE STATE="DELHI";
```

| customer_id | first_name | last_name | email             | city  | state | join_date  | is_premium |
|-------------|------------|-----------|-------------------|-------|-------|------------|------------|
| 103         | Rohan      | Gupta     | rohan.g@email.com | Delhi | Delhi | 2024-03-10 | 1          |

1 row in set (0.00 sec)

```
mysql> SELECT * FROM PRODUCTS WHERE CATEGORY IN ('ELECTRONICS','HOME');
```

| product_id | product_name         | category    | brand        | unit_price | stock_qty |
|------------|----------------------|-------------|--------------|------------|-----------|
| 201        | Wireless Earbuds     | Electronics | BoAt         | 1499.00    | 250       |
| 203        | Smart Watch          | Electronics | Noise        | 2999.00    | 150       |
| 205        | Bluetooth Speaker    | Electronics | JBL          | 3499.00    | 200       |
| 207        | Laptop Stand         | Electronics | AmazonBasics | 899.00     | 180       |
| 206        | Bedsheet Set         | Home        | Spaces       | 1299.00    | 300       |
| 208        | Cushion Covers (Set) | Home        | HomeCenter   | 599.00     | 400       |

```
mysql> SELECT *
-> FROM orders
-> WHERE order_date BETWEEN '2024-08-10'
-> AND '2024-08-25'
-> AND status != 'Cancelled';
```

| order_id | customer_id | order_date | status    | total_amount |
|----------|-------------|------------|-----------|--------------|
| 1004     | 101         | 2024-08-10 | Delivered | 3499.00      |
| 1006     | 105         | 2024-08-15 | Delivered | 5898.00      |
| 1007     | 106         | 2024-08-18 | Pending   | 1299.00      |
| 1008     | 103         | 2024-08-20 | Delivered | 899.00       |
| 1009     | 107         | 2024-08-25 | Shipped   | 6098.00      |

5 rows in set (0.22 sec)

```
mysql> SELECT *
-> FROM products
-> WHERE category='Electronics'
-> AND unit_price > 2000;
```

| product_id | product_name      | category    | brand | unit_price | stock_qty |
|------------|-------------------|-------------|-------|------------|-----------|
| 203        | Smart Watch       | Electronics | Noise | 2999.00    | 150       |
| 205        | Bluetooth Speaker | Electronics | JBL   | 3499.00    | 200       |

2 rows in set (0.04 sec)

```
mysql> SELECT *
-> FROM orders
-> WHERE total_amount > 5000;
```

| order_id | customer_id | order_date | status    | total_amount |
|----------|-------------|------------|-----------|--------------|
| 1003     | 103         | 2024-08-05 | Shipped   | 7498.00      |
| 1006     | 105         | 2024-08-15 | Delivered | 5898.00      |
| 1009     | 107         | 2024-08-25 | Shipped   | 6098.00      |

3 rows in set (0.03 sec)

```
mysql> SELECT * FROM orders WHERE status ='DELIVERED';
```

| order_id | customer_id | order_date | status    | total_amount |
|----------|-------------|------------|-----------|--------------|
| 1001     | 101         | 2024-08-01 | Delivered | 4498.00      |
| 1002     | 102         | 2024-08-03 | Delivered | 799.00       |
| 1004     | 101         | 2024-08-10 | Delivered | 3499.00      |
| 1006     | 105         | 2024-08-15 | Delivered | 5898.00      |
| 1008     | 103         | 2024-08-20 | Delivered | 899.00       |
| 1010     | 108         | 2024-08-28 | Delivered | 1598.00      |

6 rows in set (0.02 sec)

```
mysql> SELECT COUNT(*) AS TOTAL_ORDERS FROM ORDERS;
```

| TOTAL_ORDERS |
|--------------|
| 10           |

1 row in set (0.02 sec)

```
mysql> SELECT SUM(total_amount) AS TOTAL_REVENUE FROM ORDERS
-> WHERE STATUS='DELIVERED';
```

| TOTAL_REVENUE |
|---------------|
| 17191.00      |

1 row in set (0.09 sec)

```
mysql> SELECT category,AVG(unit_price) AS AVG_UNIT_PRICE
-> FROM products
-> GROUP BY category;
```

| category    | AVG_UNIT_PRICE |
|-------------|----------------|
| Clothing    | 2699.000000    |
| Electronics | 2224.000000    |
| Home        | 949.000000     |

3 rows in set (0.00 sec)

```
mysql> SELECT
->     product_id,
->     SUM(quantity) AS total_quantity
-> FROM order_items
-> GROUP BY product_id;
```

| product_id | total_quantity |
|------------|----------------|
| 201        | 3              |
| 202        | 1              |
| 203        | 2              |
| 204        | 2              |
| 205        | 2              |
| 206        | 2              |
| 207        | 2              |
| 208        | 3              |

8 rows in set (0.02 sec)

```
mysql> SELECT status,
-> COUNT(*) AS TOTAL_ORDERS,
-> SUM(total_amount) AS TOTAL_REVENUE
-> FROM orders
-> GROUP BY STATUS
-> ORDER BY total_revenue DESC;
```

| status    | TOTAL_ORDERS | TOTAL_REVENUE |
|-----------|--------------|---------------|
| Delivered | 6            | 17191.00      |
| Shipped   | 2            | 13596.00      |
| Cancelled | 1            | 2999.00       |
| Pending   | 1            | 1299.00       |

4 rows in set (0.05 sec)

```
mysql> SELECT category,
-> MAX(unit_price) AS most_expensive,
-> MIN(unit_price) AS cheapest
-> FROM products
-> GROUP BY category;
```

| category    | most_expensive | cheapest |
|-------------|----------------|----------|
| Clothing    | 4599.00        | 799.00   |
| Electronics | 3499.00        | 899.00   |
| Home        | 1299.00        | 599.00   |

3 rows in set (0.28 sec)

```
mysql> SELECT
->     category,
->     AVG(unit_price) AS avg_price
-> FROM products
-> GROUP BY category
-> HAVING AVG(unit_price) > 2000;
```

| category    | avg_price   |
|-------------|-------------|
| Clothing    | 2699.000000 |
| Electronics | 2224.000000 |

2 rows in set (0.04 sec)

```
mysql> SELECT
-> o.order_id,
-> o.order_date,
-> c.first_name,
-> c.last_name,
-> o.total_amount
-> FROM orders o
-> INNER JOIN customers c
-> ON o.customer_id = c.customer_id;
```

| order_id | order_date | first_name | last_name | total_amount |
|----------|------------|------------|-----------|--------------|
| 1001     | 2024-08-01 | Aarav      | Sharma    | 4498.00      |
| 1004     | 2024-08-10 | Aarav      | Sharma    | 3499.00      |
| 1002     | 2024-08-03 | Priya      | Patel     | 799.00       |
| 1003     | 2024-08-05 | Rohan      | Gupta     | 7498.00      |
| 1008     | 2024-08-20 | Rohan      | Gupta     | 899.00       |
| 1005     | 2024-08-12 | Sneha      | Reddy     | 2999.00      |
| 1006     | 2024-08-15 | Vikram     | Singh     | 5898.00      |
| 1007     | 2024-08-18 | Ananya     | Iyer      | 1299.00      |
| 1009     | 2024-08-25 | Karan      | Mehta     | 6098.00      |
| 1010     | 2024-08-28 | Divya      | Nair      | 1598.00      |

10 rows in set (0.05 sec)

```
mysql> SELECT
-> c.customer_id,
-> c.first_name,
-> c.last_name,
-> o.order_id,
-> o.order_date,
-> o.status,
-> o.total_amount
-> FROM customers c
-> LEFT JOIN orders o
-> ON c.customer_id = o.customer_id
-> ORDER BY c.customer_id;
```

| customer_id | first_name | last_name | order_id | order_date | status    | total_amount |
|-------------|------------|-----------|----------|------------|-----------|--------------|
| 101         | Aarav      | Sharma    | 1001     | 2024-08-01 | Delivered | 4498.00      |
| 101         | Aarav      | Sharma    | 1004     | 2024-08-10 | Delivered | 3499.00      |
| 102         | Priya      | Patel     | 1002     | 2024-08-03 | Delivered | 799.00       |
| 103         | Rohan      | Gupta     | 1003     | 2024-08-05 | Shipped   | 7498.00      |
| 103         | Rohan      | Gupta     | 1008     | 2024-08-20 | Delivered | 899.00       |
| 104         | Sneha      | Reddy     | 1005     | 2024-08-12 | Cancelled | 2999.00      |
| 105         | Vikram     | Singh     | 1006     | 2024-08-15 | Delivered | 5898.00      |
| 106         | Ananya     | Iyer      | 1007     | 2024-08-18 | Pending   | 1299.00      |
| 107         | Karan      | Mehta     | 1009     | 2024-08-25 | Shipped   | 6098.00      |
| 108         | Divya      | Nair      | 1010     | 2024-08-28 | Delivered | 1598.00      |

10 rows in set (0.06 sec)

```
mysql> SELECT
-> o.order_id,
-> p.product_name,
-> oi.quantity,
-> oi.unit_price,
-> oi.discount_pct
-> FROM orders o
-> INNER JOIN order_items oi
->     ON o.order_id = oi.order_id
-> INNER JOIN products p
->     ON oi.product_id = p.product_id
-> ORDER BY o.order_id;
```

| order_id | product_name         | quantity | unit_price | discount_pct |
|----------|----------------------|----------|------------|--------------|
| 1001     | Wireless Earbuds     | 2        | 1499.00    | 0.00         |
| 1001     | Laptop Stand         | 1        | 899.00     | 10.00        |
| 1002     | Cotton T-Shirt       | 1        | 799.00     | 0.00         |
| 1003     | Smart Watch          | 1        | 2999.00    | 0.00         |
| 1003     | Running Shoes        | 1        | 4599.00    | 5.00         |
| 1004     | Bluetooth Speaker    | 1        | 3499.00    | 0.00         |
| 1005     | Smart Watch          | 1        | 2999.00    | 0.00         |
| 1006     | Wireless Earbuds     | 1        | 1499.00    | 10.00        |
| 1006     | Running Shoes        | 1        | 4599.00    | 5.00         |
| 1007     | Bedsheet Set         | 1        | 1299.00    | 0.00         |
| 1008     | Laptop Stand         | 1        | 899.00     | 0.00         |
| 1009     | Bluetooth Speaker    | 1        | 3499.00    | 0.00         |
| 1009     | Cushion Covers (Set) | 2        | 599.00     | 15.00        |
| 1010     | Bedsheet Set         | 1        | 1299.00    | 0.00         |
| 1010     | Cushion Covers (Set) | 1        | 599.00     | 0.00         |

15 rows in set (0.02 sec)



```
mysql> SELECT
-> product_id,
-> SUM(quantity) AS qty_sold
-> FROM order_items
-> GROUP BY product_id
-> ORDER BY qty_sold DESC
-> LIMIT 5;
```

| product_id | qty_sold |
|------------|----------|
| 201        | 3        |
| 208        | 3        |
| 203        | 2        |
| 204        | 2        |
| 205        | 2        |

5 rows in set (0.08 sec)

```
mysql> SELECT
-> DATE_FORMAT(order_date, '%Y-%m') AS month,
-> COUNT(*) AS orders,
-> SUM(total_amount) AS revenue
-> FROM orders
-> GROUP BY month
-> ORDER BY month;
```

| month   | orders | revenue  |
|---------|--------|----------|
| 2024-08 | 10     | 35085.00 |

1 row in set (0.02 sec)

```
mysql> -- Top Customers
mysql> SELECT
  -> c.customer_id,
  -> CONCAT(c.first_name, ' ', c.last_name) AS customer_name,
  -> SUM(o.total_amount) AS total_spent
  -> FROM customers c
  -> JOIN orders o
  -> ON c.customer_id = o.customer_id
  -> GROUP BY c.customer_id, customer_name
  -> ORDER BY total_spent DESC
  -> LIMIT 5;
```

| customer_id | customer_name | total_spent |
|-------------|---------------|-------------|
| 103         | Rohan Gupta   | 8397.00     |
| 101         | Aarav Sharma  | 7997.00     |
| 107         | Karan Mehta   | 6098.00     |
| 105         | Vikram Singh  | 5898.00     |
| 104         | Sneha Reddy   | 2999.00     |

5 rows in set (0.06 sec)

```
mysql> -- Duplicate Orders Check
mysql> SELECT
  -> order_id,
  -> COUNT(*)
  -> FROM orders
  -> GROUP BY order_id
  -> HAVING COUNT(*) > 1;
Empty set (0.01 sec)
```

```
mysql> SELECT product_name, unit_price,
-> CASE
-> WHEN unit_price < 1000 THEN 'Budget'
-> WHEN unit_price BETWEEN 1000 AND 3000 THEN 'Mid-Range'
-> ELSE 'Premium'
-> END AS price_tier
-> FROM products;
```

| product_name         | unit_price | price_tier |
|----------------------|------------|------------|
| Wireless Earbuds     | 1499.00    | Mid-Range  |
| Cotton T-Shirt       | 799.00     | Budget     |
| Smart Watch          | 2999.00    | Mid-Range  |
| Running Shoes        | 4599.00    | Premium    |
| Bluetooth Speaker    | 3499.00    | Premium    |
| Bedsheet Set         | 1299.00    | Mid-Range  |
| Laptop Stand         | 899.00     | Budget     |
| Cushion Covers (Set) | 599.00     | Budget     |

8 rows in set (0.03 sec)

```
mysql> SELECT
-> SUM(CASE WHEN status = 'Delivered' THEN 1 ELSE 0 END) AS delivered_orders,
-> SUM(CASE WHEN status != 'Delivered' THEN 1 ELSE 0 END) AS not_delivered_orders
-> FROM orders;
```

| delivered_orders | not_delivered_orders |
|------------------|----------------------|
| 6                | 4                    |

1 row in set (0.03 sec)

```
mysql> SELECT COUNT(*) FROM customers;
```

```
+-----+
```

```
| COUNT(*) |
```

```
+-----+
```

```
|      8 |
```

```
+-----+
```

```
1 row in set (0.02 sec)
```

```
mysql> SELECT COUNT(*) FROM products;
```

```
+-----+
```

```
| COUNT(*) |
```

```
+-----+
```

```
|      8 |
```

```
+-----+
```

```
1 row in set (0.00 sec)
```

```
mysql> SELECT COUNT(*) FROM orders;
```

```
+-----+
```

```
| COUNT(*) |
```

```
+-----+
```

```
|     10 |
```

```
+-----+
```

```
1 row in set (0.00 sec)
```

```
mysql> SELECT COUNT(*) FROM order_items;
```

```
+-----+
```

```
| COUNT(*) |
```

```
+-----+
```

```
|     15 |
```

```
+-----+
```

```
1 row in set (0.02 sec)
```

```
mysql> --Missing Values \c
```

```
mysql> SELECT *
```

```
-> FROM customers
```

```
-> WHERE first_name IS NULL
```

```
-> OR email IS NULL;
```

```
Empty set (0.01 sec)
```

```
mysql> -- Invalid Entries
```

```
mysql>
```

```
mysql> SELECT *
```

```
-> FROM products
```

```
-> WHERE unit_price <= 0;
```

```
Empty set (0.03 sec)
```